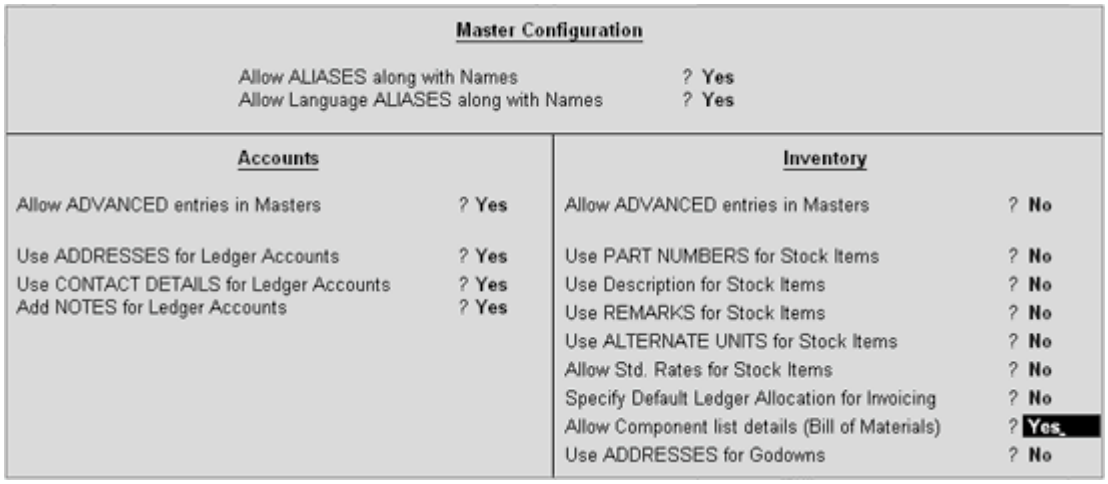
Creating Bill of Materials for a Stock Item

Bill of Material feature is useful for manufacturing organisations and can be set up through **F12: Configure**.

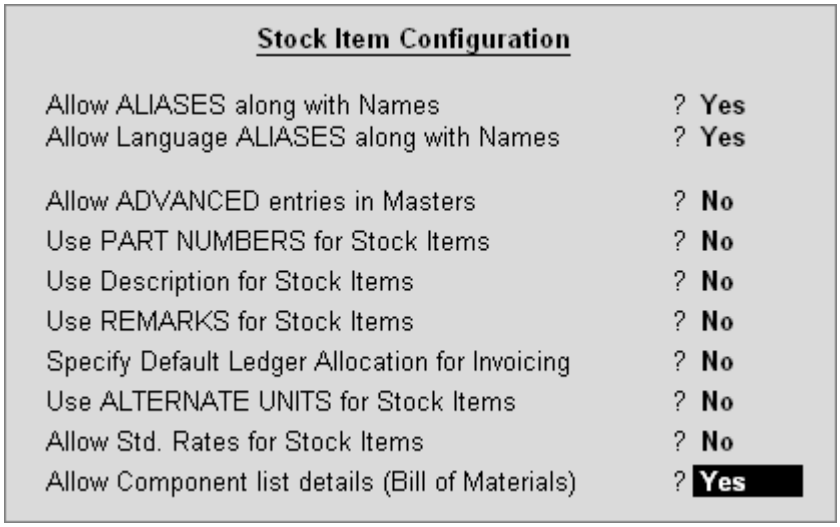
To setup Bill of Materials

Go to **Gateway of Tally > F12: Configure >** select **Accts/Inventory Info.** > set **Allow Component List Details (Bill of Materials)** to **Yes**



OR

Go to **Gateway of Tally > Inventory Info > Stock Items > Create** > Press **F12:Configure (Stock Item Configuration)** and set **Allow Component List Details (Bill of Materials)** to **Yes.**



For a Stock Item, you can create

- **A Single BOM**

- **Multiple BOM**
- **Single/Multiple BOM with Type of Item**

Creating Single Bill of Material for a Stock Item

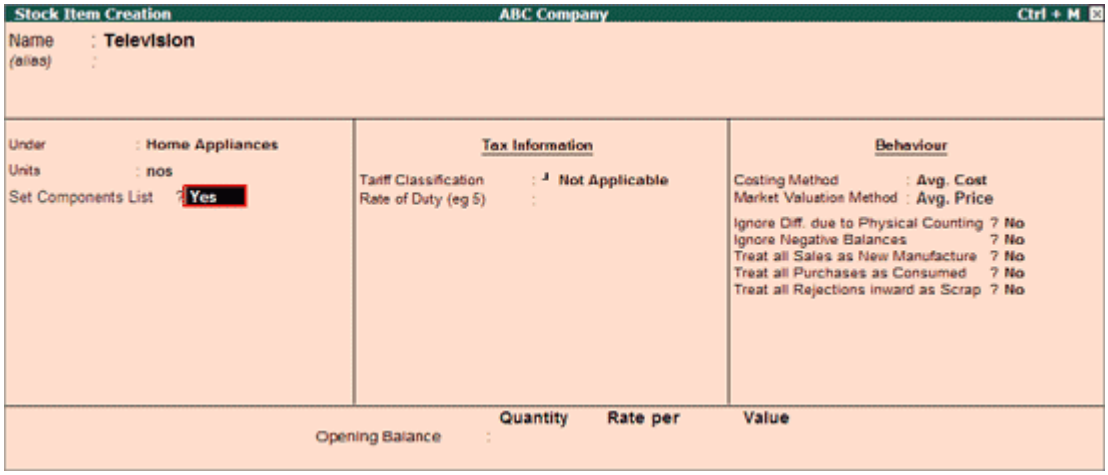
Single BOM can be created for a stock item where you use **standard set of inputs** to manufacture finished goods.

Let us create a new Stock Item Television and specify BoM.

Go to **Gateway of Tally > Inventory Info. > Stock Items > Create**

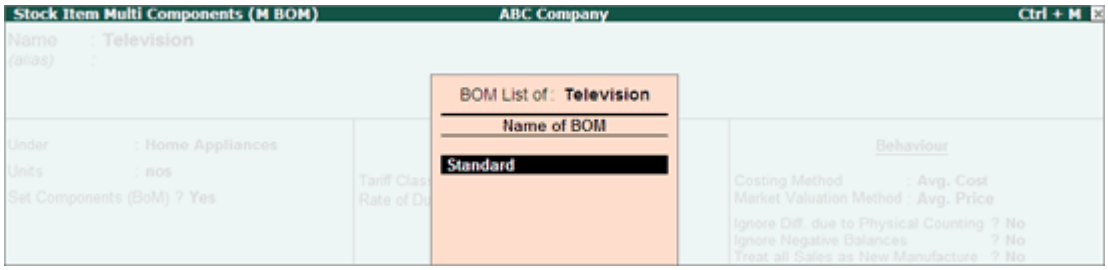
In **Stock Item Creation** screen

- Enter the **Stock Item Name**, select the appropriate **Stock Group** and specify the appropriate **Unit of Measurement** in **Unit** field
- **Set Component List**: To create the component list required to manufacture the Finished goods enable this option to **Yes**



Note: The **Set Components List** option will not displayed in Stock Item master on enabling the configuration option **Allow Component list details (Bill of Material)**,option will display in item master only on entering/ specifying the unit of measure in the **Units** field.

- Press **Enter** to view Stock Item Multi Components (M BOM) screen
- In **Stock Item Multi Components (M BOM)** screen
  - Under **Name of BOM** specify component list Name - **Standard** and press enter to view Stock Item Components (BOM) screen

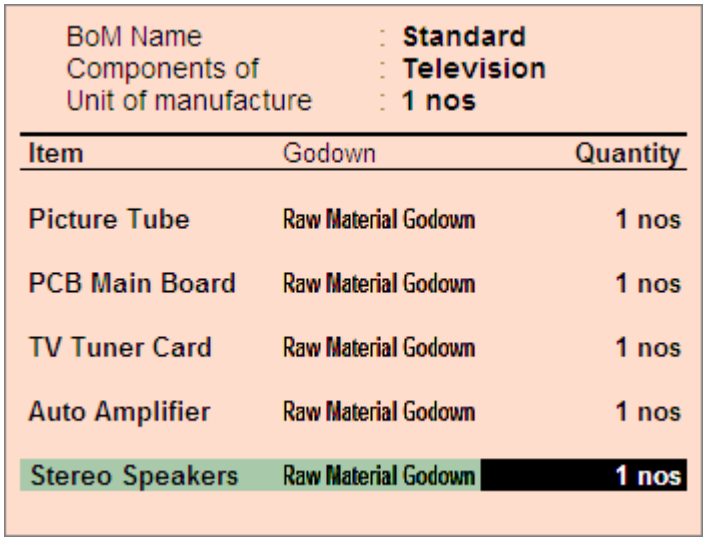


- In **Stock Item Components (BOM)** screen
    - **BOM Name**: By default the BOM Name specified in **Stock Item Multi Components (M BOM)** screen will be defaulted and application will not allow to move the cursor to BOM Name field.
    - **Components of**: In this field application displays the Name of the Stock Item for which the components are defined.
    - **Unit of manufacture**: Specify the appropriate number of finished product that will be manufactured by using the components identified in the component list.
    - **Item**: In this field identify the raw materials which are required to manufacture the finished goods
    - **Godown**: If there are multiple Godowns or Locations, specify the storage location of the items from where the items will be used for manufacture.

Note: The godown option will be displayed only when **Maintain Multiple Godown** feature is enabled in **F11: Features (Inventory Features)**

    - **Quantity**: In this field mention the quantity of Raw Material required to manufacture the finished goods.

Observe to create **1 Television** we require **5 different inputs**. The **Quantity** of inputs consumed to manufacturer finished goods is **1 each** of 5 different inputs.



BoM Name : **Standard**
Components of : **Television**
Unit of manufacture : **1 nos**

Item	Godown	Quantity
Picture Tube	Raw Material Godown	1 nos
PCB Main Board	Raw Material Godown	1 nos
TV Tuner Card	Raw Material Godown	1 nos
Auto Amplifier	Raw Material Godown	1 nos
Stereo Speakers	Raw Material Godown	1 nos

- Specify other required details in the stock item master and accept the screen.

Creating Manufacturing Journal for a Stock Item with Single Bill of Materials

All the manufacturing transactions can be accounted through a Stock Journal.

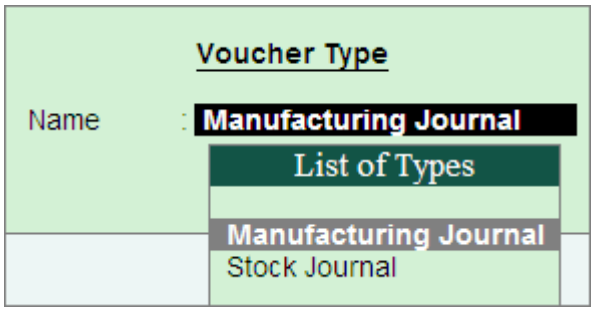
- To record manufacture of finished goods in Stock Journal - **Alter** the **Voucher Type** by **setting** the option **Use as a Manufacturing Journal** to **Yes** . **Or**
- **Create** a **Manufacturing Journal** under **Stock Journal** by setting the option **Use as a Manufacturing Journal** to **Yes**

Note: **Bill of Material** should be created before accounting the manufacture of an item.

Go to **Gateway of Tally > Inventory Vouchers**

1. Select the button **F7: Stk. Jrnl** or press **Alt + F7.**

If you have created a new voucher type **Manufacturing Journal**, a pop-up menu will be displayed with the options default Stock Journal and Manufacturing Journal.



Select **Manufacturing Journal** to view the Manufacturing Journal voucher creation screen

- **Name of Product**: Select the product which needs to be manufactured from the List of Items.
- **BoM Name**: In this field the **BoM** created for the stock item will be displayed automatically with one more option **Not Applicable**.

Note: The list of Raw materials required for the manufacture of finished goods will be displayed in **Components (Consumption)** side based on the BOM defaulted.

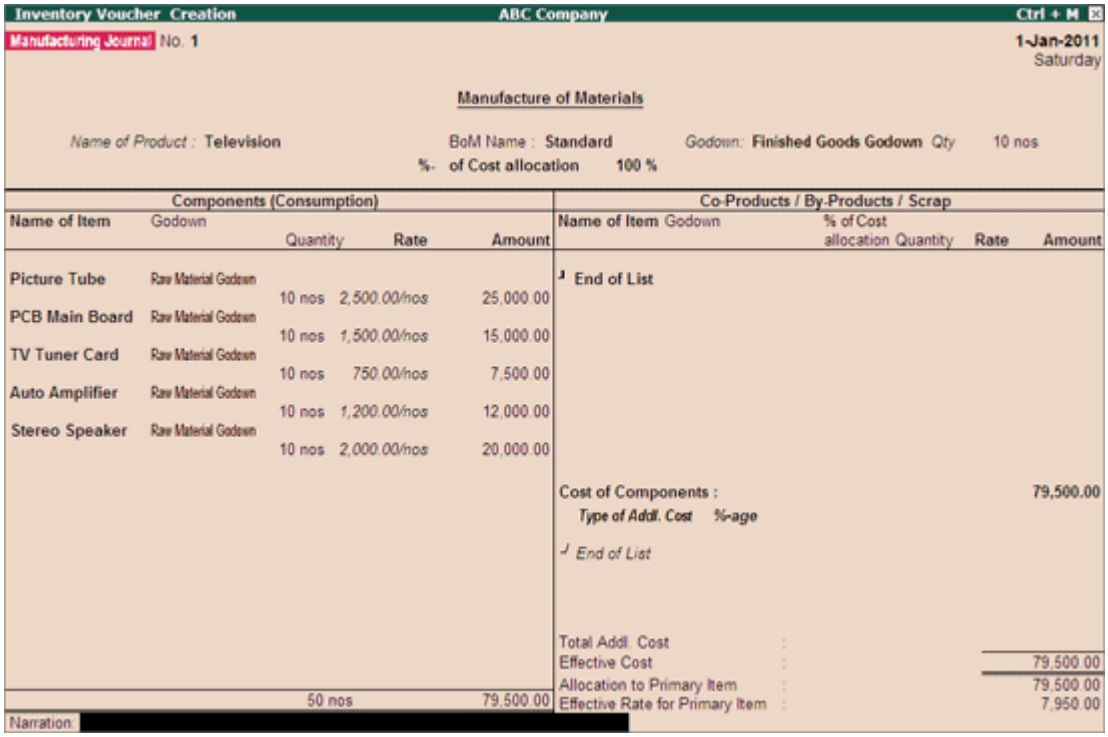
- **Godown**: Select the godown where you would like to place the produced goods.

Note: **Godown** option will be displayed only when **Maintain Multiple Godown** feature is enabled in F11: Features (Inventory Features)

- **Quantity**: In this field specify the quantity of finished goods manufactured.
- **Components (Consumption)**: In this section the components required to manufacture the finished goods will be displayed automatically based on the components defined in the BoM.

  Quantity of Raw material will be defaulted based on the **quantity requirement** defined in the **BOM** multiplied by the **Quantity** of finished goods manufactured.

The completed Manufacturing Journal is displayed as shown



Creating Multiple Bill Of Material for a Stock Item

Multiple BOM can be created for a stock items where you use varying set of components to manufacture a specific finished goods based on the availability of goods or customer requirements.

The facility to create **Multiple Bill of Materials** for Stock Item has been provided in Tally.ERP 9 Release 3.0. Where you can create **multiple BOM** with **different component list** for a stock item & appropriate BOM can be selected in the manufacturing Voucher.

This feature will be useful where you are using different components to manufacture the same stock item based on the customer requirement

Example: **Customer A** wants **Television** to be manufactured with **specific brand of Picture Tube**. Where as **other customers** has not specified any such specifications. In this scenario ABC company needs 2 BOM, one for **Customer A**as per his component specification & other may be a standard BOM for all other customers.

To create multiple Bill of Material

Go to **Gateway of Tally > Inventory Info. > Stock Items > Create**

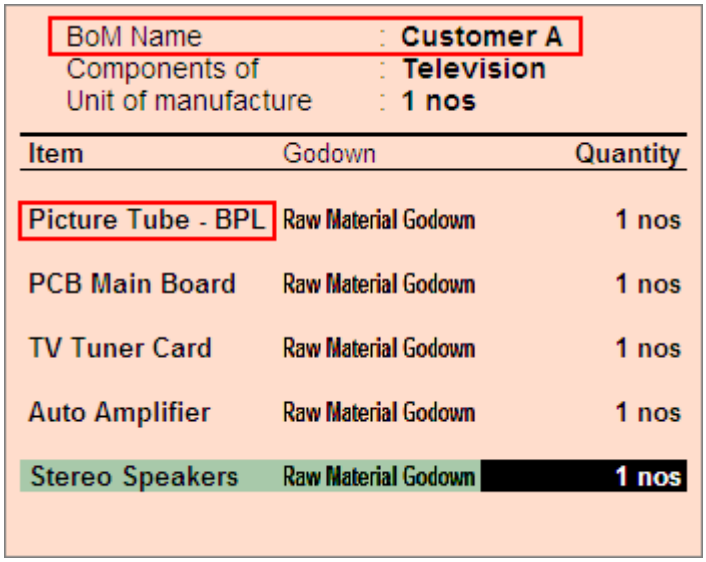
In **Stock Item Creation** screen

- Enter the **Stock Item Name**, select the appropriate **stock group** and specify the appropriate **unit of measurement** in **Unit** field
- **Set Component List**: To create the component list required to manufacture the Finished goods enable this option to **Yes** and press **Enter** to view Stock Item Multi Components (M BOM) screen
- In **Stock Item Multi Components** (M BOM) screen

- Under **Name of BOM** specify component list Name - **Customer A a**nd press enter to view **Stock Item Components (BOM)** screen

  If BOM is specifically created for a specific customer/sale you can enter the customer or sale name in Name of BOM field to identify the BOM while recording manufacturing transaction.

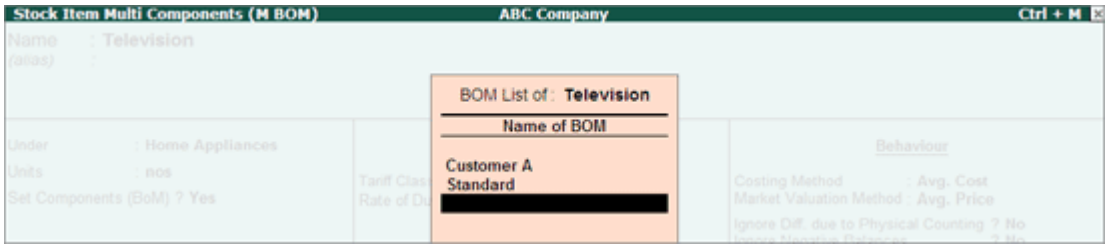
- In **Stock Item Components (BOM)** screen
  - **BOM Name**: By default the BOM Name specified in Stock Item Multi Components (M BOM) screen will be defaulted and application will not allow to move the cursor to BOM Name field.
  - **Components of**: In this field application displays the Name of the Stock Item for which the components are defined.
  - **Unit of Manufacture:** Specify the appropriate number of finished product that will be manufactured by using the components identified in the component list.
  - **Item**: In this field identify the raw materials which are required to manufacture the finished goods
  - **Godown**: If there are multiple Godowns or Locations, specify the storage location of the items from where the items will be used for manufacture.
  - **Quantity**: In this field mention the quantity of Raw Material required to manufacture the finished goods.



- Press **Enter** to save the Stock Item components for BOM - **Customer A** and to view Stock Item Multi Component screen
- In **Stock Item Multi Component** screen enter the second Bill of Material name- **Standard** in **Name of BOM** field
- In **Stock Item Components (BOM)** screen specify the **Unit of Manufacture, Item, Godown** and **Quantity** details as shown

BoM Name : **Standard**
Components of : **Television**
Unit of manufacture : **1 nos**

Item	Godown	Quantity
Picture Tube	Raw Material Godown	1 nos
PCB Main Board	Raw Material Godown	1 nos
TV Tuner Card	Raw Material Godown	1 nos
Auto Amplifier	Raw Material Godown	1 nos
Stereo Speakers	Raw Material Godown	1 nos



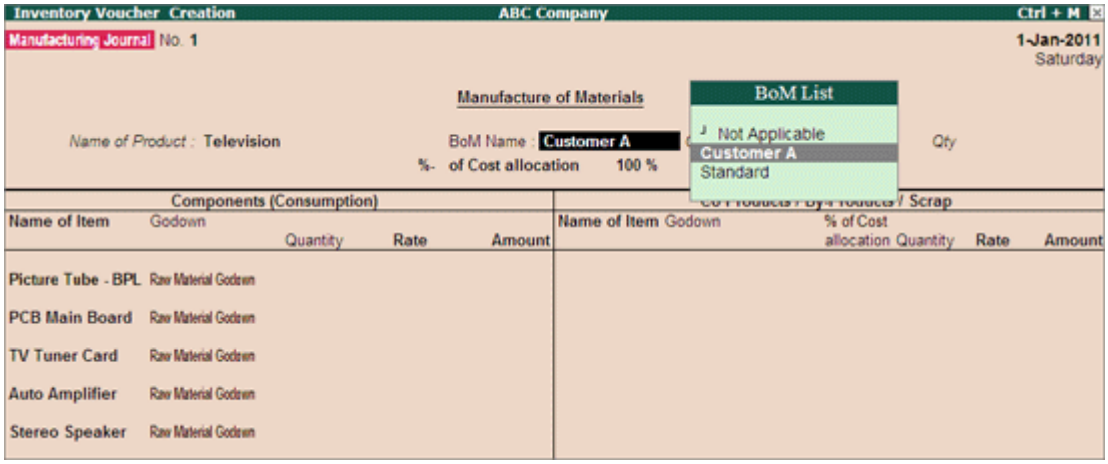
Note: In the above scenario for Stock item - **Television** we have created **2 BOM**. Similarly you can create **multiple BOM** with different component list for the stock items.

- Specify other required details in the stock item master and accept the screen.

Creating Manufacturing Journal for a Stock Item with Multiple Bill of Materials

In Manufacturing Journal voucher

- **Name of Product**: Select the product which needs to be manufactured from the List of Items.
- **BoM Name**: In this field Tally.ERP 9 displays the **BOM List** for selection of **BOM**, as multiple BOM are created for the finished goods.



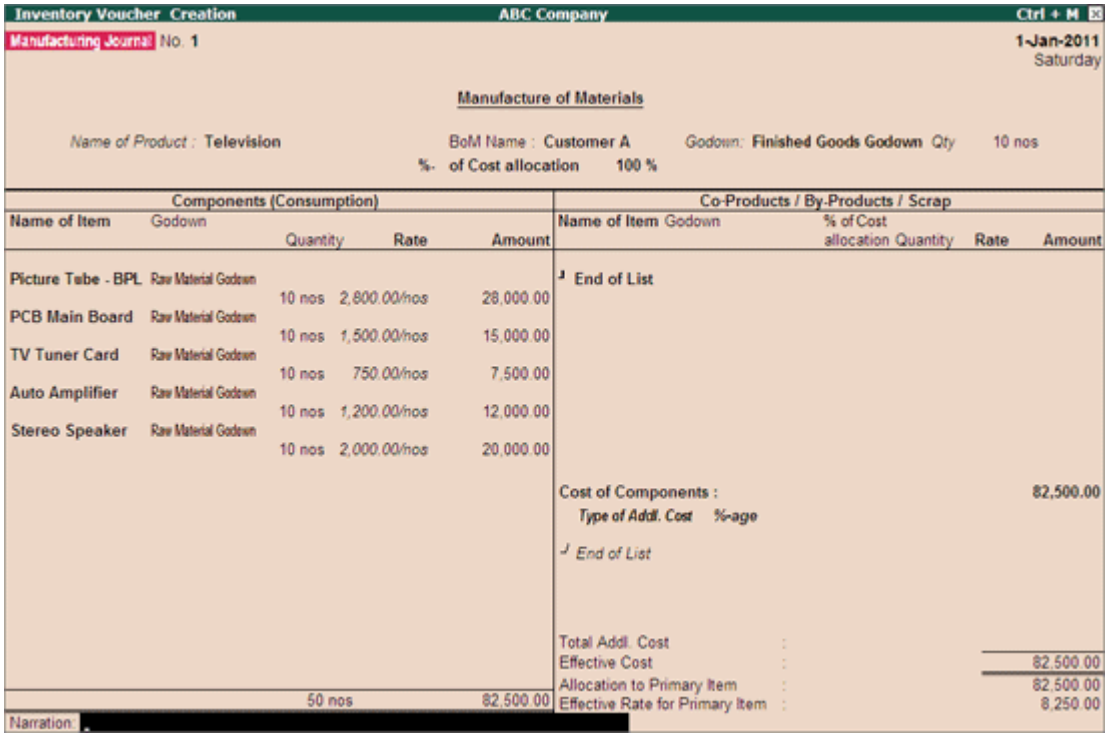
Note: The list of Raw materials required for the manufacture of finished goods will be displayed in Components (Consumption) side based on the BOM defaulted.

- **Godown**: Select the godown where you would like to place the produced goods.

Note: **Godown** option will be displayed only when **Maintain Multiple Godown** feature is enabled in F11: Features (Inventory Features)

- **Quantity**: In this field specify the quantity of finished goods manufactured.
- **Components (Consumption)**: In this section the components required to manufacture the finished goods will be displayed automatically based on the components defined in the BoM.

  Quantity of Raw material will be defaulted based on the **quantity requirement** defined in the **BOM** multiplied by the **Quantity** of finished goods manufactured.



Creating Single/Multiple Bill Of Material with Type of Item

In some of the manufacturing processes along with the finished goods, Co-Products or By-Products or Scrap may be produced.

In this scenario while defining the **BOM** for the Finished goods, you may want to identify the **inputs** which are **consumed** to manufacture Finished goods & the **By-products /Co-Products/ scrap** manufactured.

Now in Tally.ERP 9 Release 3.0, you can create the BOM by selecting & identifying the components as consumption, By-Product/Co-Product/ Scrap.

Let us take the case of manufacture of **Sugar** and create the **BOM** for the stock Item

To create Bill of Material with Type of Item

Go to **Gateway of Tally > Inventory Info. > Stock Items > Create**

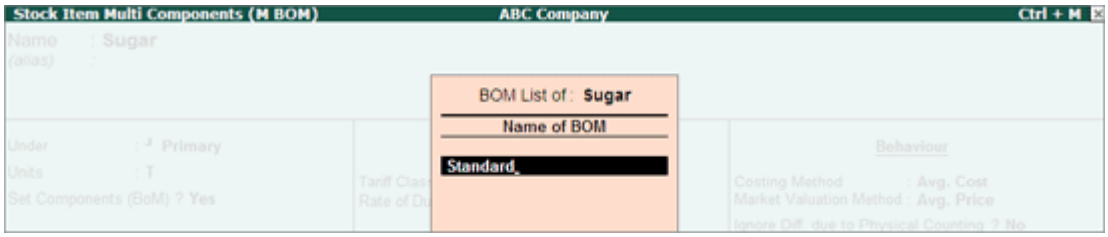
In **Stock Item Creation** screen

1. Enter the **Stock Item Name**, select the appropriate **Stock Group** and specify the appropriate **Unit of Measurement** in **Unit** field

2. **Set Component List**: To create the component list required to manufacture the Finished goods enable this option to **Yes** and press **Enter** to view Stock Item Multi Components (M BOM) screen

3. In **Stock Item Multi Components (M BOM)** screen

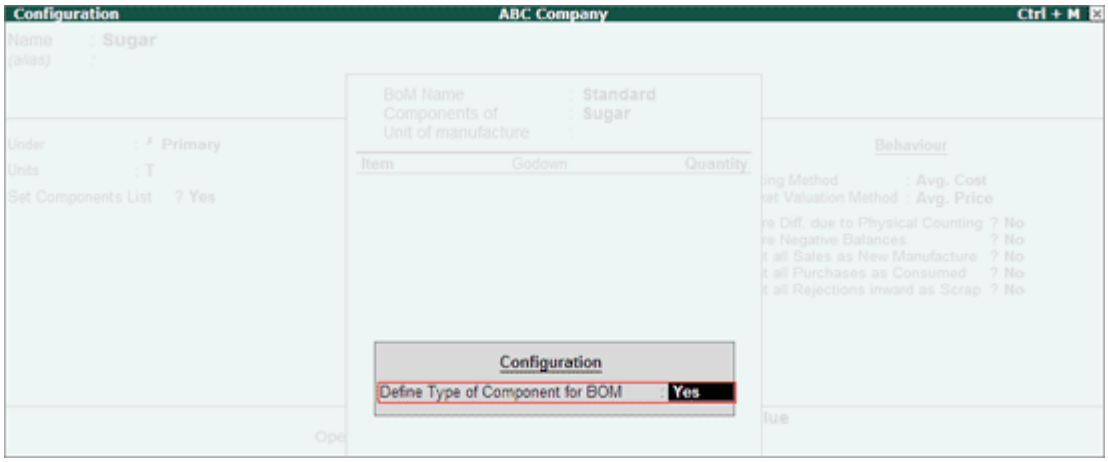
- Under **Name of BOM** specify component list Name as - **Standard**



4. In **Stock Item Components (BOM)** screen,

Set up:

- In **F12: Configuration**, enable the option **Define Type of Component for BOM** to **Yes**

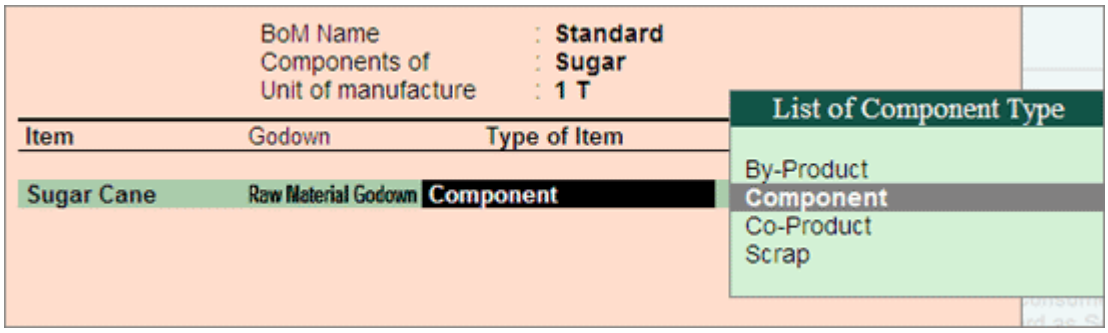


On enabling the configuration option **Define Type of Component for BOM** in **Stock Item Components (BOM)** screen a new column **Type of Item** will be displayed.

- **BOM Name**: By default the BOM Name specified in **Stock Item Multi Components (M BOM)** screen will be defaulted and application will not allow to move the cursor to BOM Name field.

- **Components of**: In this field application displays the Name of the Stock Item for which the components are defined.

- **Unit of Manufacture:** Specify the appropriate number of finished product that will be manufactured by using the components identified in the component list.
- **Item**: In this field identify the raw materials which are required to manufacture the finished goods
- **Godown**: If there are multiple Godowns or Locations, specify the storage location of the items from where the items will be used for manufacture.
- **Type of Item**: In this field select the appropriate type for the item selected.
    - **By-Product**: Select this classification for the stock items which are produced as By-Product of finished goods
    - **Component**: Select this classification for the Raw materials /inputs which are used or consumed for the production of finished goods.
    - **Co-Product**: Select this classification for the stock items which are produced as Co-product of finished goods
    - **Scrap**: Select this classification for the stock item produced as scrap in the process of manufacture of finished goods.



- **Quantity**: In this field mention the quantity of Raw Material required to manufacture the finished goods or the Co- Products/ By-Products /scrap manufactured.

Note: Cursor will not move to **Rate (%)** column when the **Component Type** selected is **Consumption**.

- **Rate (%)**: In this field you can specify the **percentage of cost of components** to be considered to determine the **cost of production of Co-products/By-Products/Scrap**.

Note: If the **Component Type** selected is **Co-product/By-Product or Scrap**, cursor will move to Rate (%) column to define the percentage of cost of production of Co-product/By-Product or Scrap.

The completed Stock Item Component (BOM) screen for Stock Item - **Sugar** appears as shown

BoM Name : **Standard**
Components of : **Sugar**
Unit of manufacture : **1 T**

Item	Godown	Type of Item	Quantity	Rate (%)
Sugar Cane	Raw Material Godown	Component	4 T	
Lime	Raw Material Godown	Component	20 KG	
Bagasse	By-Product Godown	By-Product	1 T	0.50 %
Molasses	By-Product Godown	By-Product	100 Ltrs	0.25 %

Observe to manufacture Sugar, **Sugar Cane & Lime** are identified as **Raw materials (Component)**. **Bagasse** & **Molasses** are identified as **By-Products**.

Note: When the **Manufacturing Journal** is recorded to account the manufacture of **Sugar** using the above created **BOM** the **stock items** identified as **Component** will be displayed under **Components (Consumption)** section. The **stock items** identified as **CoProduct/ByProduct** or **Scrap** will be displayed under **Co-Products/By-Products/Scrap** section

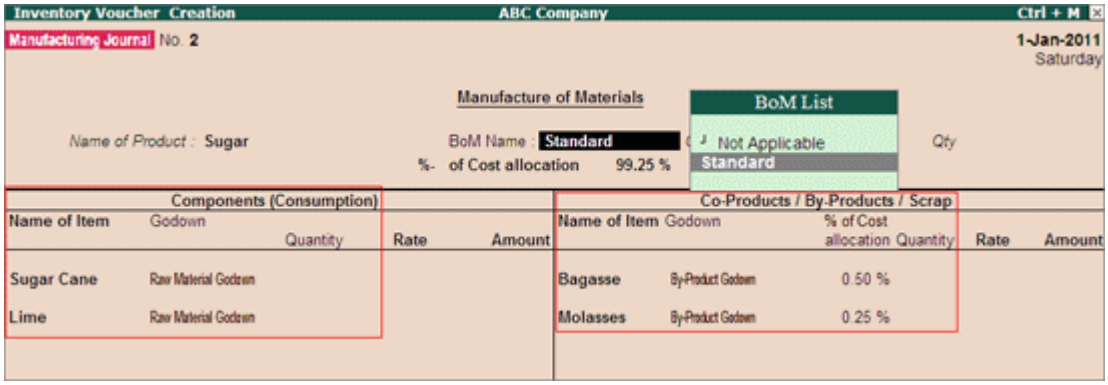
- Press **Enter** to save the Stock Item components for BOM - **Standard** and to view Stock Item Multi Component screen. If required you can create the second BOM for the stock Item
- Specify other required details in the stock item master and accept the screen.

Creating Manufacturing Journal for a Stock Item with single/Multiple BOM created with Component Type

In Manufacturing Journal voucher

- **Name of Product**: Select the product which needs to be manufactured from the List of Items.
- **BOM Name**: In this field the **BoM** created for the stock item will be displayed automatically with one more option **Not Applicable**.

  For Stock Item Sugar we have created only one BOM, hence that BOM is defaulted. If Multiple BOM are created application displays the BOM list for selection



Note:Based on the **BOM** selected

* The list of **Raw materials** required to manufacture finished goods will be displayed in **Components (Consumption)**side

*The **By-products & scrap** produced during the manufacture will be displayed under **Co-Products/By-Products/Scrap** side

The default of Raw Materials & **Co-Products/By-Products/Scrap** will be based on the **Component Type** defined in BOM.

- **Godown**: Select the godown where you would like to place the produced goods.

Note: **Godown** option will be displayed only when **Maintain Multiple Godown** feature is enabled in F11: Features (Inventory Features)

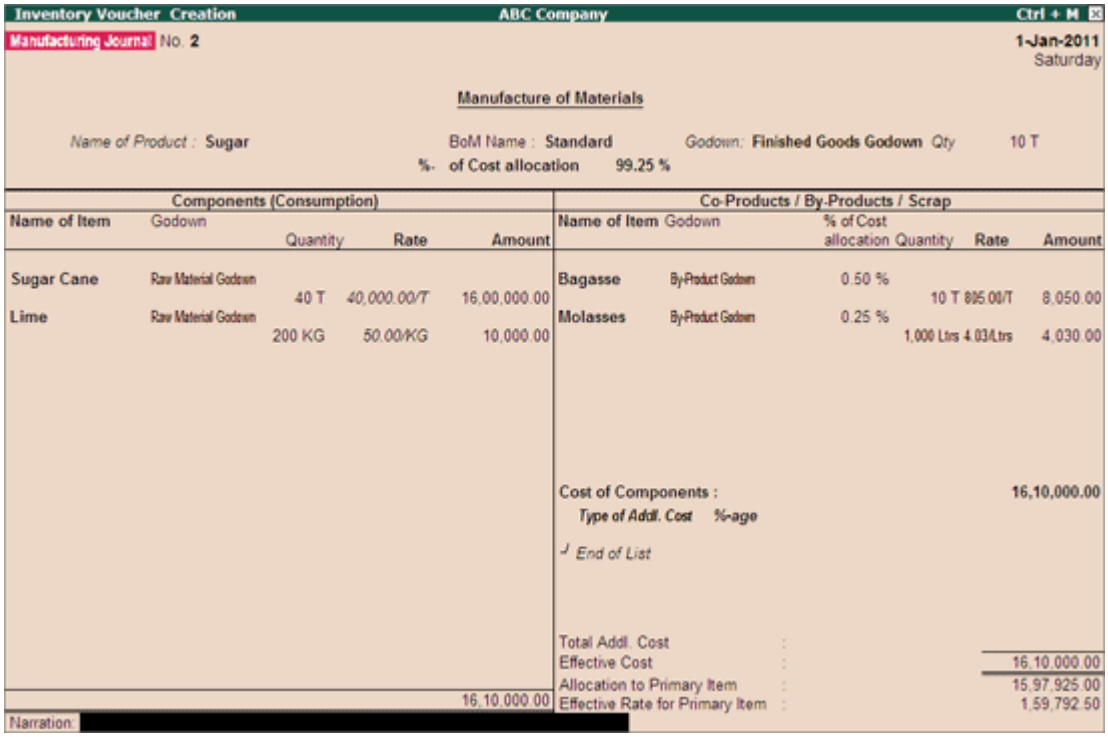
- **Quantity**: In this field specify the quantity of finished goods manufactured.
- **Components (Consumption)**: In this section the components required to manufacture the finished goods will be displayed automatically based on the components defined in BoM.

  Quantity of Raw material will be defaulted based on the **quantity requirement** defined in the **BOM** multiplied by the **Quantity** of finished goods manufactured.

- **Co-Products / By-Products / Scrap**: In this section the co-products/ by-products/scrap produced during manufacture of finished goods will be displayed automatically based on the components defined & Component Type defined in BOM

  Quantity of **Co-Products / By-Products / Scrap** will be defaulted based on the **quantity** defined in the **BOM** multiplied by the **Quantity** of finished goods manufactured.

  - **% of Cost**: In this field, the percentage specified in **Rate (%)** column for the By-products/Co-Products or scrap while creating BOM will be defaulted. The **% of Cost** will be considered to determine the **cost of production of Co-products/By-Products/Scrap**.



- For field description refer **Manufacturing Journals**.

Copy Components of BOM

In **Bill of Material** creation the capability has been enhanced to **copy** the **components** of **one BOM** to another BOM.

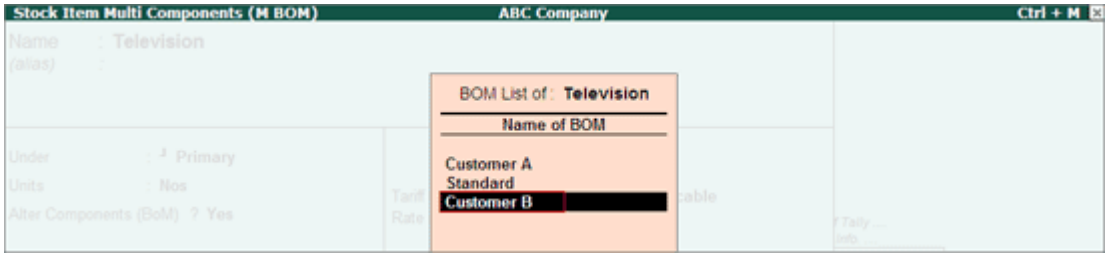
Example: For stock item Television, ABC company has 2 BOM - **Customer A** and **Standard**. Now company wants to create a new BOM - **Customer B** as Customer wants Television to be manufactured with specific brand of Picture Tube - **Picture Tube LG**. Except Picture Tube, the components used to manufacture **Television** will be same as defined in **Standard** BOM. Hence ABC Company wants to copy the components of standard BOM to Customer B.

To create the BOM

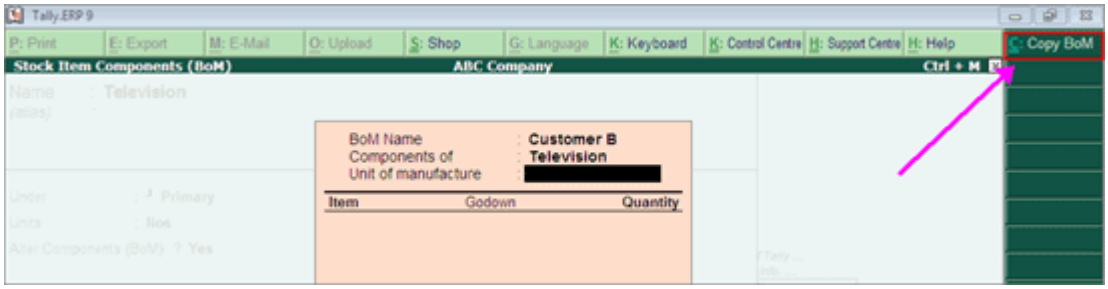
Go to **Gateway of Tally > Inventory Info. > Stock Items > Create**

In **Stock Item Alteration**

- **Alter Components (BOM)**: To create the component list required to manufacture the Finished goods enable this option to **Yes** and press **Enter** to view Stock Item Multi Components (M BOM) screen
- In **Stock Item Multi Components** (M BOM) screen
  - The Existing BOMs - **Customer A** and **Standard** are displayed (to move down use **Down Arrow Key**)
  - Enter the BOM Name - **Customer B**



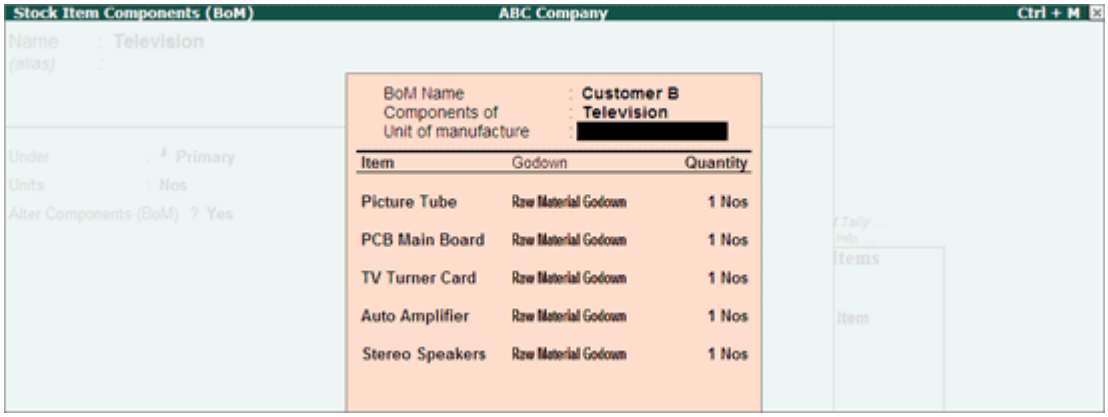
- Press **Enter** to view **Stock Item Components (BOM)** screen
- In **Stock Item Components (BOM)** screen
- To copy bill of material, **Click** on **Copy BOM button** on the Buttons Bar or press **Ctrl +C** to view **Copy BoM Details** screen



- In **Copy BoM Details** screen
  - **Copy Component for BOM**: In this field select the BOM from which the components to be copied. Here ABC company is copying the components from **Standard**.



- Press **Enter**. The components of Standard BoM will get copied to BOM - **Customer B**.



- **Unit of Manufacture:** Specify the appropriate number of finished product that will be manufactured by using the components identified in the component list.
- **Item**: In this field select the **Picture Tube - LG** and Retain other items/components.

